

TAMIL NADU STATE BOARD - STANDARD 11 CHEMISTRY VOLUME 2

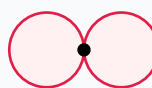
CHAPTER 15: ENVIRONMENTAL CHEMISTRY

PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 11 English Medium
Subject:	Chemistry Volume 2
Chapter Name:	Environmental Chemistry

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Atomic Orbitals Wave Functions (s & p)

p orbital (px)

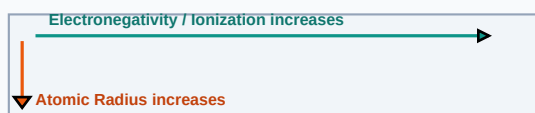
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Eutrophication	The nutrient enrichment of water bodies leading to excessive growth of algae, depleting dissolved oxygen levels.
Acid Rain	Rainwater having pH below 5.6 caused by atmospheric sulphur dioxide and nitrogen oxide emissions reacting with moisture.
Biochemical Oxygen Demand (BOD)	The amount of dissolved oxygen required by aerobic bacteria to decompose organic wastes in a water sample.

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

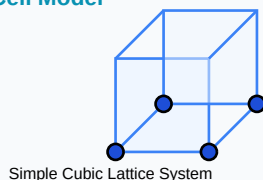
Periodic Properties Trends Direction Map



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

Crystal Solid Lattice Unit Cell Model



Essential Formulas, Laws & Identities:

- Acid Rain formation: $\text{SO}_2 + \text{O}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4$
- Ozone depletion: $\text{CF}_2\text{Cl}_2 \rightarrow \text{CF}_2\text{Cl} + \text{Cl}^* \mid \text{Cl}^* + \text{O}_3 \rightarrow \text{ClO}^* + \text{O}_2$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

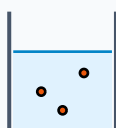
Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Explain the greenhouse effect and list the primary greenhouse gases.

Step-by-step Solution:

The trapping of solar infrared radiation by atmospheric gases, warming the planet. Primary gases: Carbon dioxide (CO_2), Methane (CH_4), Water vapor, Chlorofluorocarbons.

Titration Beaker & Solute Precipitation



Reacting Mixture

HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Describe the chemical mechanisms of ozone layer depletion in stratosphere by chlorofluorocarbons (CFCs).

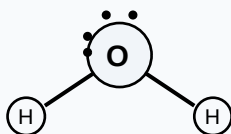
Analytical Solution Strategy:

CFCs are decomposed by UV light in stratosphere, releasing free chlorine radicals: $\text{CF}_2\text{Cl}_2 \rightarrow \text{CF}_2\text{Cl} + \text{Cl}^*$.

Chlorine radical reacts with ozone to form chlorine monoxide radical and oxygen: $\text{Cl}^* + \text{O}_3 \rightarrow \text{ClO}^* + \text{O}_2$.

ClO^* reacts with atomic oxygen: $\text{ClO}^* + \text{O} \rightarrow \text{Cl}^* + \text{O}_2$, regenerating the chlorine catalyst to destroy more ozone.

Lewis Dot Structure (H₂O Bent Molecular Geometry)



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

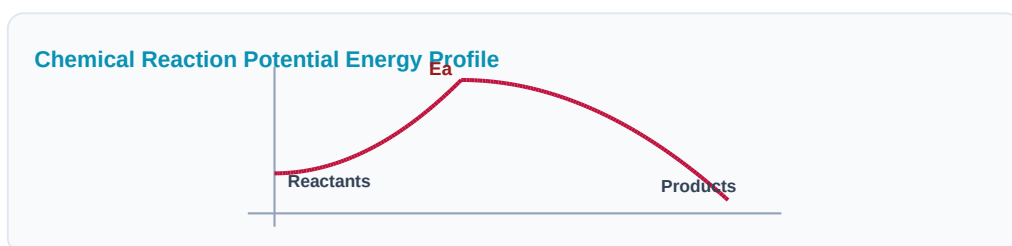
Q1 (1-Mark): Define BOD and COD.

Q2 (2-Mark): What is photochemical smog? State its key components.

Q3 (2-Mark): List the harmful effects of acid rain on marble structures.

Q4 (5-Mark): Explain how green chemistry helps reduce industrial pollution.

Q5 (5-Mark): What are primary and secondary air pollutants?



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Waste Treatment

Aeration basins reduce BOD levels of municipal sewage before discharging water.

Application Case: Industrial Synthesis

Supercritical carbon dioxide is used as a green solvent, replacing toxic organic options.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

Aromatic Benzene Ring (C₆H₆) Resonance



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Electrochemical Galvanic Cell System

